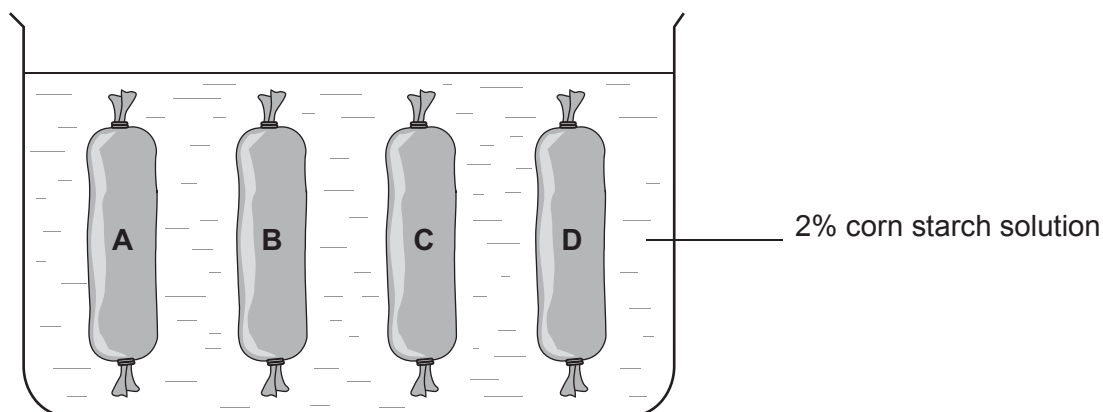


4. Starch is usually insoluble in water, but some types of starch e.g. corn starch can be treated to dissolve in water. Four pieces of Visking tubing **A**, **B**, **C** and **D**, contain equal volumes of corn starch solution of different concentrations. They are placed in a beaker containing a 2% corn starch solution and left for 45 minutes.



- (a) Describe, in terms of molecules, the structure of starch. [2]

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- (b) After 45 minutes:

- the volume of **A** is unchanged
- the volume of **B** has increased
- the volume of **C** has decreased
- the volume of **D** has increased more than **B**

- (i) Explain which tube contained a corn starch solution of lower concentration than that in the beaker. [2]

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- (ii) Explain why tube **A** must have contained the 2% corn starch solution. [2]

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